THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET B

Chapter 10: **Tissues in Action** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The correct order of organisation in a multicellular organism is: 1

(a) tissue → cell → organ → system **(b)** organ → tissue → cell → system
(c) cell → tissue → organ → organ system **(d)** cell → organ → tissue → system

2. Meristematic cells are characterised by: 1

(a) wide intercellular spaces **(b)** large vacuoles
(c) thick cell walls **(d)** dense cytoplasm and a prominent nucleus

3. Pores in the leaf epidermis that allow gas exchange are: 1

(a) stomata **(b)** lenticels **(c)** vessels **(d)** root hair

4. Parenchyma with air spaces helping aquatic plants float is called: 1

(a) collenchyma **(b)** sclerenchyma **(c)** aerenchyma **(d)** chlorenchyma

5. Water and minerals are conducted upward in plants by: 1

(a) xylem **(b)** phloem **(c)** cambium **(d)** epidermis

6. Cork is formed by the activity of the: 1

(a) apical meristem **(b)** vascular cambium
(c) cork cambium **(d)** intercalary meristem

7. Bones are connected to muscles by: 1

(a) tendons **(b)** nerves **(c)** ligaments **(d)** cartilage

8. Muscles of the heart are: 1

(a) skeletal and involuntary **(b)** smooth and voluntary
(c) cardiac and involuntary **(d)** skeletal and voluntary

9. The shoulder joint is a: 1

(a) fixed joint **(b)** pivot joint
(c) ball and socket joint **(d)** hinge joint

10. The number of pairs of ribs in the human rib cage is: 1

(a) 14 **(b)** 24 **(c)** 12 **(d)** 10

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What are annual rings? What do they indicate? 2

12. What is differentiation? 2

13. State two functions of the epidermis in plants. 2

14. What is transpiration? State its importance. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Compare parenchyma, collenchyma and sclerenchyma. 3

16. Why can the stem of a young mango sapling bend while a coconut husk is tough? 3

17. Describe the structure of xylem. 3

OR

Describe the structure and function of phloem. 3

18. What would happen if a plant could not transport food from leaves to roots? 3

OR

Why is plant xylem called ‘dead tissue doing the work’? Explain how water rises against gravity. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Coconut husk fibres make tough mats; a young sapling stem bends easily. 5

(i) Name the tissue in the husk.
(ii) Name the tissue in the sapling stem.
(iii) Compare the two.
(iv) What would happen if the sapling had only the husk tissue?

OR

(i) Describe the structure of parenchyma, collenchyma and sclerenchyma. 5
(ii) Give one function of each.

20. (i) Compare xylem and phloem in a table of four points. 5
(ii) How does water rise to the top of tall trees?

OR

Case study: A farmer notices that a plant's leaves wilt in hot afternoons but recover at night. 5

(i) Which process loses water?
(ii) Which tissue replaces the water?
(iii) Which cells control the loss?
(iv) Why does the plant recover at night?

END OF QUESTION PAPER — Total: 40 Marks

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